

Noor-ul-Huda Siddiqui

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SUMMARY

Digital IC Design and Verification Engineer with a Computer Science background and hands-on experience in RTL development, FPGA implementation, functional verification, and simulation-based debugging. Worked on hardware and software projects involving UART communication, cache controller design, VGA framebuffer timing, AMBA bus protocol implementation, and an interactive cache simulator built in C++/Qt6. Strong practical exposure to Verilog/SystemVerilog, ModelSim/Questasim, Xilinx Vivado, Zybo and Arty A7 FPGA boards, and constraint-based hardware verification. Currently expanding into ASIC physical design and learning the RTL-to-GDS flow using OpenLane 2.0.

PROJECTS

Interactive Cache Visualizer & Simulator

- Built a desktop application to simulate cache behavior across direct-mapped, fully associative, and set-associative configurations.
- Added support for address breakdown, cache line tracking, replacement strategy handling, and live hit/miss statistics.
- Designed a Qt6-based interface for configuring cache settings, loading trace inputs, and observing cache activity visually.

Technologies: C++17 , Qt6 , Qt Widgets , CMake , object-oriented design

[github link](#)

Synchronous UART with Baud Rate Controller

- Developed a synchronous UART transmitter and receiver with baud-rate generation, parity support, and loopback testing.
- Used a clock-enable approach to keep the design in one clock domain and improve timing cleanliness.
- Verified the design through simulation and board-level testing on the Zybo FPGA platform.

Technologies: Verilog HDL, ModelSim/Questasim, Zybo (Zynq-7000), XDC constraints, FPGA verification

[github link](#)

VGA Controller with SRAM Framebuffer

- Created a VGA timing controller for 640x480 @ 60Hz output with sync pulse generation and framebuffer-driven pixel display.
- Implemented clock generation, video timing logic, and SRAM-based pixel fetching for stable display output.
- Verified timing and output behavior through simulation and FPGA-oriented implementation on the Arty A7-100T board.

Technologies: Verilog HDL, PLL-based clocking , SRAM framebuffer , ModelSim , F4PGA , Arty A7-100T

[github link](#)

4-Way Set-Associative Cache Controller

- Implemented a 4-way set-associative cache controller with parallel tag comparison, valid/dirty bit tracking, LRU replacement, and write-back logic.
- Structured the controller around FSM states for cache lookup, hit processing, miss handling, refill, and eviction.
- Validated cache operation using testbenches and waveform analysis, with FPGA targeting for the Arty A7-100T.

Technologies: Verilog HDL, FSM design, SRAM interface, ModelSim, F4PGA, Arty A7-100T

[github link](#)

AMBA Bus Protocol Work

- Implemented and studied the AMBA APB protocol for peripheral communication.
- Simulated and analyzed AHB and AXI concepts to strengthen understanding of SoC interconnects and high-performance bus behavior.
- Verified protocol behavior through RTL modeling and simulation-based checks.

Technologies: Verilog, SystemVerilog, bus protocols, RTL verification

OpenLane 2.0 RTL-to-GDS

- Currently studying full ASIC implementation flow using OpenLane 2.0.
- Building hands-on understanding of synthesis, floorplanning, PDN generation, IO placement, CTS, routing, timing analysis, and physical verification.

Technologies: Verilog, SystemVerilog, bus protocols, RTL verification

EXPERIENCE

Digital IC Design & Verification Trainee

May 2026 – Present

Pakistan Software Export Board (PSEB)

- Building a strong foundation in RTL design, computer architecture, and ASIC methodology.
- Developing and verifying Verilog/SystemVerilog designs for UART, cache controllers, SRAM interfacing, and VGA timing.
- Working on AMBA APB concepts and exploring AHB and AXI interconnects.
- Designed a 4-way set-associative cache controller and a VGA controller with SRAM framebuffer support.
- Built an interactive cache simulator in C++ and Qt6.
- Practicing simulation, waveform analysis, and FPGA-based verification.
- Currently learning RTL-to-GDS flow with OpenLane 2.0, including synthesis, floorplanning, placement, routing, and STA.

EDUCATION

Bachelor of Science in Computer Science

2022 - 2026

UIT University, Karachi

- CGPA : 3.01/4.00

Intermediate in Computer Science

2021 - 2022

Govt. Degree College SRE Majeed Stadium Road, Karachi

- GRADE : A

Schooling

2019 - 2020

Montessori Complex High School

- GRADE : A+

SKILLS

Digital Design & Verification: RTL Design, FPGA Design, ASIC Design Flow, Physical Design, OpenLane 2.0, Static Timing Analysis (STA), APB Protocol

Tools & Platforms: Xilinx Vivado, QuestaSim, Ubuntu Linux, Git, Visual Studio Code, Visual Studio, Cisco Packet Tracer, MATLAB, Google Colab

Programming Languages: C, C++, Python, Verilog HDL, SystemVerilog, Java, JavaScript, TypeScript, C#, SQL

Web & App Development: HTML, CSS, React.js, Next.js, Node.js, Flutter

Productivity & Analytics: MS Office, Power BI, Oracle